

CloudFormation Console Deployment Guide

How to deploy JSON templates from the AWS Management Console

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1) Learning Goal

By the end of this lab, students will be able to deploy, monitor, update, and delete CloudFormation stacks directly from the AWS Console.

2) Files Used in This Lab

- 00-all-in-one.json (single-stack starter)
- 01-vpc-network.json (network layer)
- 02-ec2-instance.json (compute layer)
- 03-s3-bucket.json (storage layer)

3) Before You Start

- Sign in to AWS Console with an account that can use CloudFormation, EC2, VPC, and S3.
- Select one AWS Region and keep all stacks in the same region.
- Create (or confirm) an EC2 Key Pair in that region (for EC2 SSH access).
- Have template files downloaded locally on your computer.

4) Deploy First Stack from Console (All-in-One)

Step-by-step

- Open AWS Console and search for 'CloudFormation'.
- Click 'Create stack' > 'With new resources (standard)'.
- In 'Specify template', choose 'Upload a template file'.
- Upload: 00-all-in-one.json.
- Click Next.
- Stack name: student-all-in-one.
- Fill parameters:
 - EnvironmentName = dev
 - KeyPairName = your existing key pair
 - InstanceType = t3.micro
 - BucketNamePrefix = travel237-allinone (or your custom prefix)
- Click Next > Next > Create stack.

Monitor Progress

- In stack Events tab, wait until status becomes CREATE_COMPLETE.
- If it fails, review the first red failure event for root cause.

5) Validate Deployed Resources

- Open Outputs tab in the stack and copy values like InstancePublicIp and BucketName.
- Go to EC2 > Instances and confirm instance is running.
- Open the instance public IP in a browser; you should see the demo page.
- Go to S3 > Buckets and verify the bucket exists.
- Check bucket properties: Versioning = Enabled, Encryption = Enabled.

6) Deploy Modular Stacks from Console

A) Network Stack

- Create stack with 01-vpc-network.json.
- Use stack name: student-network.

- Set EnvironmentName=dev and desired CIDR ranges.

B) EC2 Stack (imports network outputs)

- Create stack with 02-ec2-instance.json.
- Use stack name: student-ec2.
- Set EnvironmentExportPrefix=dev and KeyPairName=your key pair.

C) S3 Stack

- Create stack with 03-s3-bucket.json.
- Use stack name: student-s3.
- Set BucketNamePrefix to a globally unique value.

7) How to Update a Stack in Console

- Open CloudFormation > select your stack.
- Click Update.
- Choose 'Replace current template' and upload updated JSON (if needed).
- Adjust parameters and click Next > Next > Update stack.
- Wait for UPDATE_COMPLETE.

8) How to Troubleshoot Common Console Errors

- Template format error: validate JSON syntax (missing comma/bracket).
- S3 bucket name already exists: change BucketNamePrefix.
- Key pair not found: choose key pair in the correct region.
- Insufficient permissions: use IAM user/role with required permissions.
- Resource limit exceeded: check service quotas in that region.

9) How to Delete Stacks Safely

Delete stacks in dependency order for modular setup:

- Delete student-ec2 first
- Delete student-s3 second
- Delete student-network last

For all-in-one, delete student-all-in-one directly.

10) Classroom Exercise

- Task 1: Deploy 00-all-in-one.json from console.
- Task 2: Validate Outputs and test EC2 public web page.
- Task 3: Delete stack and redeploy with a different EnvironmentName.
- Task 4: Deploy modular stacks and compare architecture with all-in-one.

Prepared for Unixcloudtrainings by Valery